A

Expert-defined question

Can a high-risk literature record be converted into a bounded source-data audit?

Retraction-associated record → **Public FASTQ** → **Source audit**

Public accession
SRP508771 / PRJNA1111407 / SRR29088443

Expert-defined boundary
check files and QC outputs;
do not reassess the biological claim

B

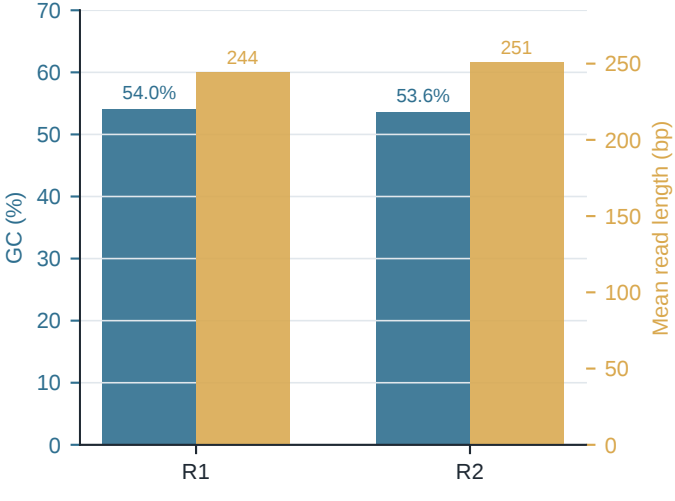
Public record vs PaleoRigor output

Read, base, and compressed-file-size values compared with the public record.

Metric	Public	PaleoRigor	Difference
R1 reads	53,571	**53,571**	**0**
R2 reads	53,571	**53,571**	**0**
Total bases	26,517,645	**26,517,645**	**0**
R1 file size (bytes)	1,240,071	**1,240,071**	**0**
R2 file size (bytes)	1,428,410	**1,428,410**	**0**

All five metadata checks matched

C



Malformed records: 0; maximum N-fraction: 4×10^{-6}

D

Permitted conclusion

Scope set by the expert; PaleoRigor did not detect the retraction itself.

Supported by the files
- read, base, and file-size agreement
- paired-end structure
- descriptive QC values
- QC report provenance

Not supported
- byte-level file identity
- the retracted conclusion
- ancient authenticity
- contamination source
- biological mechanism

Result: file-level checks matched; biological interpretation left to the expert